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Sentiment Analysis of Bengali Movie Reviews

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***Abstract*—Sentiment analysis (SA) is necessary for understanding user opinions and impacting decision-making in domains like review of products, feedback, and social media monitoring of social media . In this study, we have focused on Bangla sentiment classification using a dataset which contains 10,800 reviews, we have catagorized them into positive(0) and negative(1) sentiments. We applied BanglaBERT, mBERT, XLM-RoBERTa (XLM-R), and Gemini, we also fine-tuned them by using LoRA (Low-Rank Adaptation) and PEFT (Parameter-Efficient Fine-Tuning) techniques. we have evaluated these models based on accuracy, precision, recall, and F1-score. Among these all model, BanglaBERT given the highest performance by achieving 97.90**

***Index Terms*—Sentiment Analysis, Bengali NLP, BanglaBERT, mBERT, XLM-RoBERTa, LoRA, Text Classification, Deep Learning.**

I. Introduction

Introduction: In today's internet-based communication environment, social media platforms and other e-commerce-based websites produce numerous amounts of text data. It is quite challenging to extract meaningful information from this data to understand users' behavior, preferences. Sentiment analysis, termed opinion mining, is a Natural Language Processing (NLP) task classified as positive, negative, or neutral. It has a great range of applications, and by taking feedback from customers they predict the market flow.

Though there are notable developments in sentiment analysis for widely used languages like English, there are not many studies on languages like Bangla due to the lack of availability of datasets, or maybe because of the lack of effective computational resources (Hoque et al., 2024)11†source. More than 230 million people speak in Bengali, it is worlds 6th most spoken language, that makes sentiment analysis in Bengali crucial for businesses, researchers, and policymakers (Bhowmik et al., 2022)10†source. However, Sentiment analysis in the Bengali language leads to challenges like use of informal language, dialect variation and the presence of code-mixed text (Tareq et al., 2023)11†source. Modern innovations in deep learning and transformer-based models have added a new dimension to sentiment analysis. Traditional approaches of machine learning approaches such as Support Vector Machines (SVM) and Naïve Bayes classifiers Often faces issues with complex language arrangements and rational understanding. With the creation of transformer based models like BERT, XLM-R, and multilingual BERT (mBERT), sentiment analysis models have achieved contemporary performance in text classification tasks (Sharma Nakrani, 2024)8†source. These models have reached a new height in achieving contextual dependencies that lead to great accuracy towards sentiment predictions, especially in low-resource languages like bangla. In this study, we have worked on sentiment predictions of movie reviews using a dataset that consists of 11,808 samples. The dataset includes user sentiments labeled as either positive (0) or negative (1), featuring a mixed range of opinions. We have implemented several transformer-based models, including BanglaBERT, mBERT, XLM-R, , Gemini to evaluate their effectiveness in classifying Bengali movie reviews. Our main goal of this research was to overcome the differences in Bengali sentiment analysis by Exploiting transformer models to improve classification accuracy and provide a benchmark for future studies in this domain.

The rest of the paper structure as follows: Section II represents a review of related works on Bengali sentiment analysis. Section III tells about the dataset and preprocessing techniques that we have used. Section IV discussed the experimental set up. Section V comparatively analyses the results obtained from different models. Finally, Section VI gives a conclusion of the paper including key findings and future research options.